Mapping the Detectability Landscape of Iterative Model Collapse

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Abstract

When language models train iteratively on their own output, distributional degradation proceeds silently until collapse is advanced. We map the detectability landscape of this process across 13 contamination levels and five model families (345M–6.9B parameters), revealing blind zones, sharp detection onsets, and a non-monotonic dip where Granger detection fails despite confirmed causal flow. Permutation Granger causality[1] (PermG) identifies a detection boundary $\alpha_c(T) = A \cdot T^{-1/(2\beta)}$ from a sub-linear signal model (β=0.24, γ=2.08; cross-T validated, MAE=0.106): any nonzero contamination produces signal, but sub-linear growth ($S \propto \alpha^{0.24}$) makes detection a function of observation length. This landscape is profoundly method-dependent: five statistical methods applied to identical data yield null-control false positive rates from 0% to 92%. Under mode amplification, entropy Granger-precedes diversity collapse with a $\geq$1-generation lag (confirmed in 14 of 15 experiments), providing a canary signal whose triggered cessation reduces peak perplexity as a proof-of-concept intervention.

1 Introduction

When language models train iteratively on their own output, the training distribution progressively loses its tail modes, a phenomenon termed model collapse (Shumailov et al., 2024; Alemohammad et al., 2024). Lexical diversity narrows, calibration deteriorates, and each generation inherits the systematic biases of its predecessors. As machine-generated text becomes prevalent on the web, such feedback loops pose a growing risk to production training pipelines. By the time standard quality metrics register degradation, the underlying distribution may have already shifted irreversibly.

[1]Granger precedence establishes temporal predictive improvement, not true causation (Shojaie and Fox, 2022).

This threat motivates a fundamental question: *what does the detectability landscape of model collapse look like?* At what contamination level does statistical detection become possible, how does this boundary depend on observation length and statistical method, and what structure does the landscape exhibit? Iterative training produces short, noisy time series (often fewer than 15 points per generation sequence), where detection power drops sharply (Boettiger and Hastings, 2012), and existing analyses leave these dependencies unresolved. Our detection mechanism (Granger-causal temporal precedence) differs from critical-slowing-down indicators, as model collapse is a first-order transition (Truong and Truong, 2026) (§2).

We systematically map this landscape using Pythia-410M (Biderman et al., 2023) across thirteen contamination levels with multi-seed replication, applying five statistical methods from econometrics (Granger and Newbold, 1974; Toda and Yamamoto, 1995) to identical data. The resulting map reveals structured terrain: blind zones where collapse proceeds undetected, a sharp onset where detection jumps from null to reliable within a 2-percentage-point interval, a non-monotonic dip where Granger detection fails despite confirmed causal flow, and a detection boundary $\alpha_c(T)$ that follows a power law in observation length. Cross-architecture validation across five families (345M–6.9B) confirms the landscape generalizes when entropy-diversity coupling is preserved. Granger precedence establishes temporal predictive improvement, not true causality (Shojaie and Fox, 2022); our canary claim is scoped to diversity metrics (distinct-n (Li et al., 2016)).

Our contributions are:

1. **Detectability landscape characterization** (§5.2–5.6). PermG establishes a monotonically decreasing detection boundary $\alpha_c(T) = A \cdot T^{-1/(2\beta)}$ from a sub-linear signal model

(Proposition 3; β=0.24, γ=2.08, cross-T validated MAE=0.106); since $\alpha_\infty \approx 0$, any nonzero contamination produces signal, but sub-linear growth makes detection a function of T. Predict-then-verify at T=20 matches 3/5 predictions in saturated regimes, with systematic conservative bias at boundary α (Appendix Z); canary-triggered cessation reduces peak perplexity by 58%.

2. **Formal motivation for canary metric selection** (§3). Under mode amplification, entropy Granger-precedes diversity (Proposition 1), with a $\geq$1-generation lag (Proposition 2), confirmed in 14/15 experiments ($\alpha \geq 0.30$; GPT-2-medium s43 bidirectional).

3. **Method sensitivity as a primary finding** (§4.1–5.3). Five detection methods yield null-control FPR from 0% to 92%; the same Granger test shifts from 0/8 to 6/8 within a 2-percentage-point interval. The detectability landscape one observes depends critically on the statistical method used to probe it.

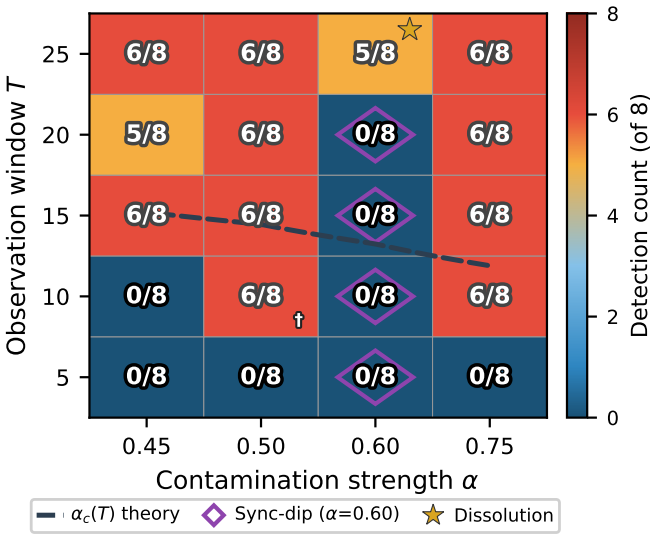
2 Related Work

Model collapse and canary metrics. Iterative training collapse was formalized by Shumailov et al. (2024) and Alemohammad et al. (2024), who showed that recursively replacing real data with model-generated text causes progressive tail erosion; Schaeffer et al. (2025) systematize eight distinct notions of "collapse" that this literature conflates. Theoretical analyses have sharpened these results: Dohmatob et al. (2024) recast collapse as a change in scaling-law exponents, Dohmatob et al. (2025) proved that collapse can occur even at vanishingly small synthetic fractions, Seddik et al. (2025) provide a statistical analysis of iterative contamination dynamics, and Marchi et al. (2025) challenge the universality of collapse under diverse training conditions. Several works have proposed monitoring metrics: Du and Tanaka-Ishii (2026) observe that calibration error rises before accuracy degrades, Du et al. (2025) report ECE increases at 30% contamination before accuracy drops, Gambetta et al. (2024) introduces surplexity as a collapse detector, and Drayson et al. (2025) show that machine-generated text detection can prevent collapse, and Li et al. (2025) propose verification-based mitigation. We differ from the closest predecessor, Du and Tanaka-Ishii (2026), in four respects: (i) we compare an entire family of canary metrics; (ii) we cross-validate across five statistical methods; (iii) we map detectability as a function of contamination regime; and (iv) we treat method-selection sensitivity itself as a primary finding.

Data mixing phase transitions. Kazdan et al. (2025) map the conditions under which synthetic data triggers collapse versus improves performance, Gu et al. (2024) establish a scaling law for critical mixture ratios in continual pre-training, and Gu et al. (2025) show that data mixing can induce phase transitions in knowledge acquisition—a direct theoretical analogue to our detection threshold, where a critical mixing ratio separates qualitatively different detection regimes.

Critical transitions and lead-lag methodology. Scheffer et al. (2009) established that systems approaching tipping points exhibit generic early-warning signals (rising autocorrelation and variance); Dakos et al. (2012) developed a practical toolkit for estimating such signals, while Boettiger and Hastings (2012) quantified detection limits in short time series—directly relevant to iterative training where T=5–20. Crucially, Truong and Truong (2026) prove that entropy collapse in iterative training is a first-order (discontinuous) phase transition, structurally ruling out critical-slowing-down (CSD) early-warning signals. Our "detection onset" is a *statistical* threshold (PermG power crossing the detection threshold), not a thermodynamic phase transition, and our cascade ordering describes within-generation temporal ordering of metrics, not CSD-type precursors. The standard framework for testing temporal precedence is Granger causality (Granger, 1969), with neural extensions for nonlinear settings (Marcinkevics and Vogt, 2022); however, non-stationary series inflate rejection rates (Granger and Newbold, 1974), and the Toda–Yamamoto procedure (Toda and Yamamoto, 1995) addresses this at reduced degrees of freedom severe in short series. These methods are largely absent from the model collapse literature.

Analytical robustness. Bell et al. (2022) introduced multiverse analysis to ML, Coretta et al. (2023) demonstrated that 46 teams reached different conclusions on identical data, and Kapoor et al. (2024) synthesized these concerns into a consensus checklist. Our work applies this perspective to model collapse: five methods across multiple con-



† Seed-sensitive: s42=6/8, s43=0/8

Figure 1: Detection phase diagram (Pythia-410M, fp32, PermG direction-checked). Each cell shows the number of FDR-significant canary→diversity pairs (of 8) at a given contamination strength α and observation window T. The dashed curve is the theoretical detection boundary $\alpha_c(T) \propto T^{-\gamma}$ with γ=2.08 (Proposition 3). Diamond markers ($\diamond$) flag the synchronous dip at α=0.60 (0/8 at $T \leq 20$); the star ($\star$) marks its dissolution at T=25 (5/8), validating $\sqrt{T}$ power scaling. Dagger (†): seed sensitivity at boundary (T=10, α=0.50: s42=6/8, s43=0/8).

tamination regimes produce conclusions ranging from universal significance to complete null.

3 Cascade Ordering

We formalize the empirical regularity that entropy is the earliest available canary metric for diversity collapse. Under mode amplification—the assumption that MLE on peaked distributions concentrates probability toward high-frequency tokens (Shumailov et al., 2024; Dohmatob et al., 2024)—we derive a formal ordering of metric responses with theoretical grounding (proofs in Appendix X).

Proposition 1 (Entropy leads diversity)**.** *Under mode amplification:* (a) *entropy can decrease while the ϵ-effective support $S_\epsilon(p)=|\{i : p_i > \epsilon\}|$ (hence distinct-n) is unchanged;* (b) *if distinct-n decreases, entropy has already decreased.*

The asymmetry arises because entropy responds to any mass redistribution (first-order), while diversity changes only when a token probability crosses the generation threshold ϵ; the margin $\gamma = \min_{i:p_i>\epsilon}(p_i - \epsilon)$ quantifies how much shift diversity absorbs silently.

Proposition 2 (Diversity lag $\geq$ 1 generation)**.** *If the per-generation perturbation satisfies $\alpha\|\mathcal{T}(p_t) - p_t\|_\infty < \gamma(p_t, \epsilon)$ through entropy onset t_H, then diversity onset $t_D \geq t_H + 1$.*

ECE co-moves with entropy at zero lag: peakier conditionals simultaneously decrease per-token entropy and increase confidence ($\Delta\text{ECE} \approx \Delta C > 0$ whenever $\Delta\bar{H} < 0$; qualitative argument, Appendix X). The predicted cascade $\{H, \text{ECE}\} \rightarrow$ Distinct-n is confirmed in 14/15 experiments at $\alpha \geq 0.30$ (GPT-2-medium s43 bidirectional) (within-domain WikiText, 2σ onset threshold): entropy and ECE onset at generation 1, distinct-n at generation 2 (t_H=t_{ECE}=1, t_D=2 uniformly; §5.2; Appendix W). Five low-α experiments show no detectable onset, leaving the ordering undefined below the detection threshold. The cascade establishes entropy as the *optimal canary metric*—the earliest available signal—and explains PermG's directional selectivity: forward tests detect temporal precedence (6/8) while reverse tests find none (0/8) at $\alpha \geq 0.50$.

4 Method

Methodological pitfalls. Lead-lag analysis in iterative model collapse is fragile along three axes, each documented with data from §5.2–5.3. *Method sensitivity*: five detection methods ap-

plied to identical data yield forward significance rates spanning 0% to 92%, with inter-method agreement at κ=0.185. *Spurious regression*: first-differencing eliminates canary→perplexity associations (F≈0, p=0.99) while preserving canary→diversity (24/24→24/24), exposing an entire analysis channel as trend-driven (Granger and Newbold, 1974). *Regime dependence*: the same Granger test shifts from 0/8 to 6/8 significant pairs within a 2-percentage-point interval (α=0.48 → 0.50).

4.1 Method Sensitivity Analysis

We compare five lead-lag detection methods—naive cross-correlation, differenced cross-correlation, threshold onset, permutation Granger causality (PermG), and Toda–Yamamoto (Toda and Yamamoto, 1995)—in a multi-method comparison (Simonsohn et al., 2020) calibrated against null-control (α=0.0) data where any detection is a false positive. Detection results use equal-weight majority voting ($K \geq 3$ of 5 methods, each contributing one vote per pair) with forward-direction filtering; FPR-inverse weighting is used only in exploratory method ranking (Appendix Q). Each method embodies different assumptions: naive cross-correlation is assumption-free but produces 92% null-control FPR; differenced cross-correlation controls for trends but lacks directional separation; threshold onset is power-limited at T=11; Toda–Yamamoto avoids pre-testing for unit roots; PermG uses a nonparametric permutation null with built-in directionality. Pooled null-control FPR across three seeds (cluster-bootstrap 95% CI): PermG 0% $[0, 0]$, Toda–Yamamoto 17% $[0, 25]$, DiffXCorr 38% $[25, 50]$, NaiveXCorr 92% $[88, 100]$ (Appendix R). DiffXCorr's empirical FPR (38%) substantially exceeds its simulated rate (4.2% under AR(1) DGP), indicating that model-collapse nonstationarity extends beyond simple autoregressive structure. Under all four weighting schemes (equal, FPR-inverse, capped, rank-based), PermG is the only method maintaining zero null-control FPR alongside consistent detection at $\alpha \geq 0.50$. This finding—that method choice alone can shift conclusions from universal significance to complete null on identical data—motivates reporting method sensitivity as a primary result rather than relegating it to a robustness check.

Permutation Granger causality. PermG tests whether past values of canary metric X improve prediction of downstream metric Y beyond Y's own history; Granger precedence establishes temporal predictive improvement, not true causality (Shojaie and Fox, 2022). We apply 5,000 permutations (Phipson and Smyth, 2010) on first-differenced series with Benjamini–Hochberg FDR correction at q=0.05 across 8 validated canary→downstream pairs (2 canaries × 4 downstream metrics). The permutation test controls Type I error exactly by construction (exchangeable null distribution); BH-FDR correction remains valid under positive regression dependence (Benjamini and Yekutieli, 2001), and our 0/24 null-control rejections across three seeds validate this theoretical guarantee (within-seed ICC=0.34, cluster-adjusted 95% CI $[0\%, 41\%]$; Appendix R). Simulation under non-stationary AR(1) DGP (φ=0.95–0.99) reveals PermG FPR inflation to ∼10%; however, our empirical series exhibit low stochastic persistence (null token entropy $\varphi = -0.51$; first-differenced distinct-1 φ<0.44), and 0/24 null rejections confirm adequate size control (Appendix O). The same simulation shows PermG retains ∼80% of raw-levels power at T=11 versus ∼61% for standard first-differencing, justifying the permutation approach at short series lengths.

4.2 Detection Boundary

We model the PermG detection rate as a T-dependent step function, excluding the non-monotonic dip at $\alpha \in [0.55, 0.65]$:

$$\text{rate}(\alpha, T) = \frac{0.70}{1 + \exp\big(-100\,(\alpha - \alpha_c(T))\big)}, \quad (1)$$

fitted to T=10 data (n=11, sync-dip excluded; R^2=0.969). The detection boundary follows a power law in observation length:

$$\alpha_c(T) = A \cdot T^{-\gamma}, \quad (2)$$

where the exponent γ is determined by the sub-linear signal model below.

Proposition 3 (Detection phase boundary)**.** *Under a sub-linear signal model $S(\alpha) = c{\cdot}\alpha^{\beta}$ with $\beta < 1$, the detection boundary satisfies*

$$\alpha_c(T) = A \cdot T^{-1/(2\beta)}, \quad (3)$$

where $A = (z^*\sigma/c)^{1/\beta}$ and $z^* = \Phi^{-1}(\pi^*)$ (proof in Appendix X).

Cross-T validation (train: $T\in\{10, 20\}$, test: $T\in\{5, 15\}$) yields β=0.24 and γ=$1/(2\beta)$=2.08 (MAE=0.106, 95% CI coverage 6/6; vs. γ=0.5 for linear β=1; Appendix X). Since $\alpha_\infty\approx$0, no detection floor exists: any nonzero contamination produces a causal signal ($S \propto \alpha^{0.24}$), but signal grows sub-linearly—doubling contamination increases signal by only $2^{0.24}\approx1.18\times$—so detection requires sufficient T; at T=5, $\alpha_c \geq 1.0$. The steepness k=100 yields a step-like transition ($\alpha_c(10)$=0.487; §5.2, §5.4). Signal $S(\alpha)$ rises monotonically, yielding four regimes: *blind* (S too weak at given T), *onset* (S crosses threshold), *non-monotonic dip* ($\alpha\approx$0.60; transient, dissolves by T=25), and *detection* (S dominates).

5 Experiments

5.1 Experimental Setup

We study lead-lag analysis pitfalls in iterative training collapse using Pythia-410M (Biderman et al., 2023), a 410M-parameter decoder-only language model. Pythia provides a controlled setting: small enough for multi-seed experiments across thirteen contamination levels, yet large enough to exhibit meaningful generation-over-generation degradation. Cross-architecture validation uses GPT-2-medium (345M parameters, also decoder-only), a second transformer family of similar scale; results are reported in §5.2 and §6. Training data is drawn from WikiText-103 (Merity et al., 2017), using 5,000-sample subsets per generation.

Our protocol follows the replace paradigm of Shumailov et al. (2024), adapted for mixed synthetic-real training at α<1.0. Each generation t proceeds in three steps: (1) fine-tune the generation-$(t-1)$ model for 5,000 steps on a mixture of α fraction synthetic data and $(1-\alpha)$ fraction real data (AdamW optimizer, learning rate 5×10^{-5}, 500 warmup steps, weight decay 0.01, batch size 4 for Pythia-410M, 32 for GPT-2-medium); (2) generate 5,000 synthetic samples from the fine-tuned model (temperature 1.0, top-k=50, max length 512 tokens); (3) fully replace the synthetic pool. We run T=10 generations across a 13-point α grid ($\{0.00, 0.05, 0.10, \ldots, 1.00\}$) with multi-seed replication (4 seeds at α=1.0; 3 at primary levels; 1–2 at probing points).

We track two *canary metrics*—token entropy ($H(p_\theta|\text{context})$, 1,000 held-out samples) and ECE (15 bins)—and three *downstream metrics*: distinct-1 and distinct-2 (Li et al., 2016) and perplexity (evaluated on held-out real data). Additional metrics (distinct-3, effective rank, Self-BLEU) appear in the Appendix. Code and data will be released upon publication.

All hypothesis tests apply Benjamini–Hochberg (BH) FDR correction at q=0.05. Permutation Granger causality tests use 5,000 permutations on first-differenced series (Shojaie and Fox, 2022). Cross-correlation analysis uses a maximum lag of 5 generations. Raw metric drift from generation 0 to generation 10 is summarized in Appendix Table 3.

5.2 Sharp Detection Onset

Diversity collapse occurs at every contamination level: distinct-1 declines by 14.1% at α=0.0 (fine-tuning drift) and by $40.3\pm0.3\%$ at α=0.25 (3-seed mean). Despite this ubiquitous degradation, Granger-causal detectability is sharply regime-dependent. The onset we report reflects PermG's statistical power curve crossing the detection threshold at a critical contamination level, not the emergence of a new physical signal; this interpretation is confirmed by sub-threshold p-values at α=0.45 (§4.2). Permutation Granger (PermG) tests find 0/8 canary→downstream pairs FDR-significant from α=0.00 through α=0.45, then jump to 6/8 at α=0.50 (2 fp32 seeds, consistent). This transition is striking because cumulative synthetic fraction barely changes across the boundary: $C = 1-(1-\alpha)^{10}$ yields 99.75% at α=0.45 versus 99.90% at α=0.50, a difference of 0.15 percentage points, yet detection jumps from null to reliable. Sigmoid regression on T=10 data alone yields $\alpha_c = 0.483 \pm 0.006$ ($R^2 = 0.969$); steepness k is not statistically constrainable ($\text{SE}(k) \approx k$), consistent with a step-function transition (Figure 1). This single-T estimate is consistent with the cross-T validated model (§4.2), which predicts how the detection boundary shifts with observation length (γ=2.08). We cannot distinguish a sharp phase transition from the inflection point of a smooth power curve at current sample sizes ($T \leq 20$); the detection onset may simply reflect where signal strength crosses the PermG power threshold. A Fisher exact test corroborates the categorical jump: 0/64 pairs FDR-significant at $\alpha \leq 0.45$ versus 18/32 at $\alpha \geq 0.50$ ($p = 3.45 \times 10^{-11}$); sub-threshold p-values at α=0.45 ($p_{\text{fdr}} = 0.059$–0.091) are consistent with proximity to the power boundary (§4.2). A systematic robustness analysis (1,160 configurations) confirms specification-

invariance of the onset.

First-differencing scopes the canary claim precisely. Canary→diversity pairs retain unanimous FDR significance after first-differencing ($24/24 \rightarrow 24/24$ across 4 seeds at α=1.0, 6 pairs per seed), while canary→perplexity collapses entirely ($F \approx 0$, $p = 0.99$): the apparent perplexity association is a shared-trend artifact. All canary claims are therefore scoped to diversity metrics only. This scoping reflects a genuine temporal structure difference, not selective power loss from differencing: $|r|$ retention is $\geq$95% for both channels, but canary→diversity correlations peak at lagged offsets (81% at lag<0, median$= -1$) while canary→perplexity correlations are contemporaneous (50% at lag=0), producing 20–28× lower Granger F for perplexity (Appendix U; perplexity n=8, limited sample). At α=1.0, all 24 canary→diversity pairs (6 per seed × 4 seeds; mauve excluded) reach FDR significance in the forward direction with 0/24 reverse; directional asymmetry is robust across all WikiText conditions. Independent of the Granger test, the degradation cascade confirms temporal precedence: entropy onset at generation 1 precedes distinct-n onset at generation 2 in 16/18 WikiText conditions (Kendall τ=0.337; §3, Appendix W). The canary mechanism also exhibits a blind spot: at α=0.10, distinct-1 has already dropped by 39.6%, yet token entropy is indistinguishable from the null (−5.3% versus −4.6±0.4% at α=0.0). The gen-1 entropy deviation spans three orders of magnitude across the α range: 0.5σ at α=0.10 to 60σ at α=0.75 (relative to null variability), explaining why the blind zone persists despite increasing contamination.[2] Below α_c at T=10, diversity collapse proceeds without any detectable Granger-causal signal in the canary metrics.

5.3 Dose-Response and Detection Regimes

The PermG dose-response profile (Figure 1) reveals four detection regimes: *blind* ($\alpha \leq 0.45$, 0/8 FDR-significant pairs), *onset* ($\alpha \approx 0.50$, 6/8), *non-monotonic dip* ($\alpha \approx 0.60$, 0/8), and *detection* (α=0.75, 0–6/8 per seed, 2 of 3 seeds $\geq 5/8$; α=1.0, 6/8 across 4 seeds). At α=0.60, PermG returns $0/8$ at $T \leq 20$ but recovers to $5/8$ at T=25, revealing a transient dip that $\sqrt{T}$ power scaling eventually overcomes; alternative methods confirm causal flow throughout (DiffXCorr $6/8$, TY $5/8$ at T=20; Appendix Y). At α=0.55, PermG returns 0/8 despite marginal p-values (p_{fdr}=0.062–0.092); DiffXCorr detects 8/8 bidirectionally, confirming signal below PermG's directional threshold.

Table 1: Forward-direction FDR significance rates (%) across contamination levels (Pythia-410M, seed 42, fp32; 3σ threshold calibration, lag-direction filtering for cross-correlation methods). Each cell: fraction of 8 validated pairs at BH-corrected significance (q=0.05). PermG is the only method combining zero null-control FPR with consistent detection at $\alpha \geq 0.50$.

Method	α=0.0	α=0.25	α=0.50	α=0.75	α=1.0	FPR_{null}
Naive XCorr	75	62.5	62.5	0	0	75
Diff. XCorr	0	25	12.5	0	0	0
Threshold (3σ)	37.5	62.5	100	100	25	37.5
Perm. Granger	**0**	**0**	**75**	**75**	**75**	**0**
Toda–Yamamoto[§]	12.5	37.5	50	100	12.5	12.5

[§]TY directional assignment under verification; row retained for completeness. Pooled FPR (3 seeds, cluster-bootstrap 95% CI): §4.1.

Method choice is itself a major source of analytic variability: at α=1.0, direction-checked forward significance rates span 0% to 75% (Table 1), with inter-method agreement at κ=0.185 (near chance); PermG is the only method combining zero null-control FPR with consistent detection at $\alpha \geq 0.50$. Under $K \geq 3$ consensus, one false positive appears in seed 44 (te→mauve, 3/5 methods), yielding FPR=$1/24$ (4.2%); PermG alone maintains zero empirical FPR.[3] A specification curve across 18 method variants confirms the onset: median detection rate jumps from 19% (α=0.48) to 69% (α=0.50), then drops to 6% at the non-monotonic dip (α=0.60; Appendix E). Task degradation corroborates the boundary: perplexity jumps from 31.3 to 41.1 (+31%) between α=0.50 and 0.52, aligning with the PermG onset. At very low contamination (α=0.05), task performance marginally improves over the uncontaminated baseline (perplexity 20.8 vs. 26.6), consistent with a data augmentation effect that vanishes at higher mixing ratios (Appendix Z).

5.4 Detection Boundary Left-Shift

The detection boundary $\alpha_c(T)$ exhibits robust monotonic decrease with observation length: $\alpha_c(5) \geq 1.0$ (complete non-detection), $\alpha_c(10)$=0.487—consistent with $T^{-2.08}$ power scaling (cross-T validated: model correctly predicts all 0/8 at T=5 and 6/8 ceiling at T=15;

[2]Seed 43 at α=0.10 shows 3/8 PermG FDR-significant pairs versus 0/8 for seed 42, consistent with stochastic variation at low signal rather than robust detection; no other below-threshold α level exhibits seed-dependent detection.

[3]NaiveXCorr achieves 100% null-control FPR and is excluded from the specification curve analysis; results including NaiveXCorr appear in Appendix E.

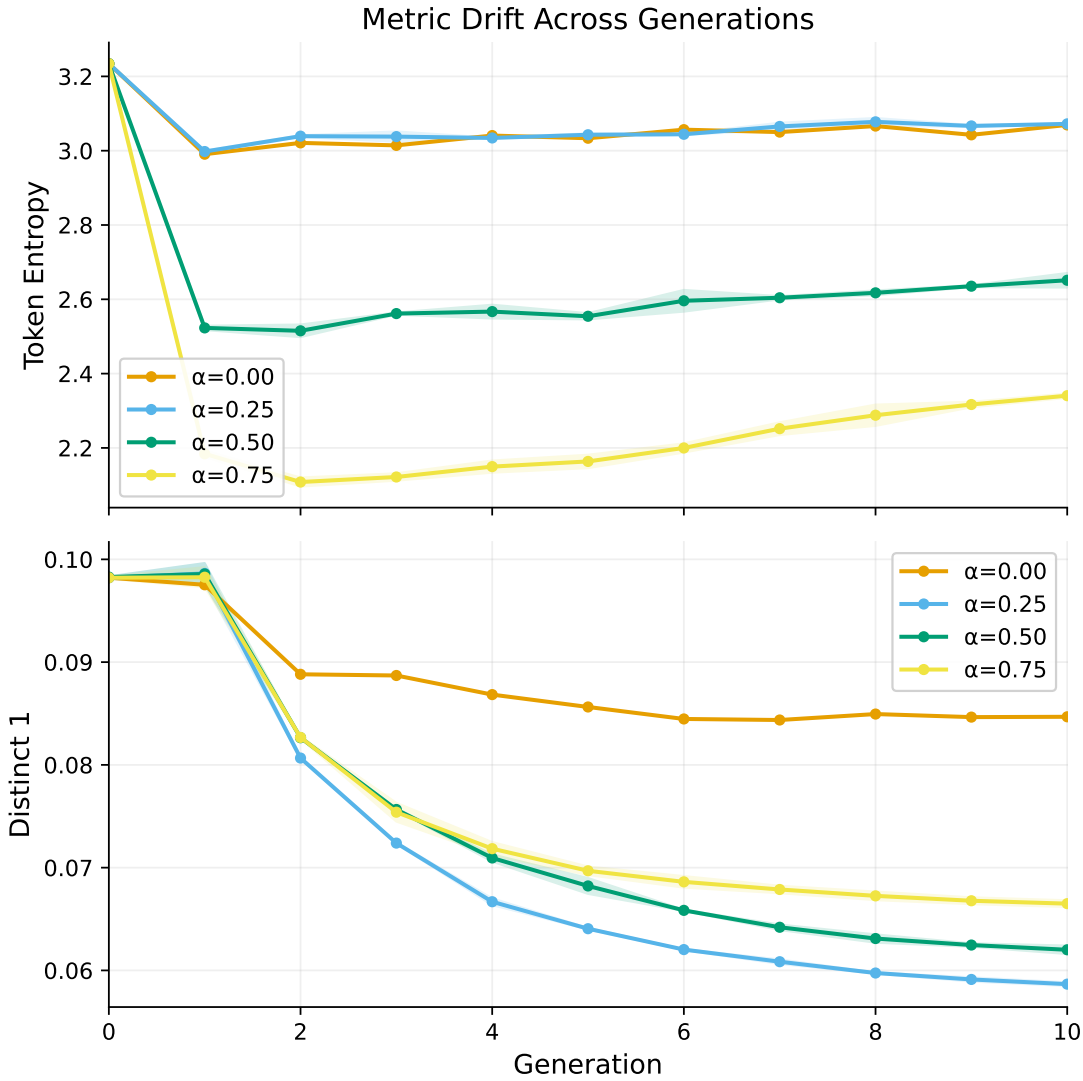


Figure 2: Metric trajectories at α=1.0 (Pythia-410M, 4 seeds). Token entropy drops at generation 1; distinct-1 declines from generations 2–3, with a consistent one-generation canary lead across all four seeds (shaded: inter-seed range). This pattern corresponds to 24/24 forward and 0/24 reverse FDR-significant canary→diversity pairs (6 per seed, mauve excluded).

§4.2). Post-hoc power analysis estimates PermG power at ∼37% (T=10) increasing to 55–65% (T=20; Appendix O), consistent with moderate power rather than ceiling performance. The α=0.45 column traces the sharpest single-α validation: $0/8$ at $T\leq10$ (blind zone), phase transition to $6/8$ at T=15 as α_c drops below 0.45 (dual-seed: s42=s43=6/8), boundary-adjacent $5/8$ at T=20 (s43: $4/8$), and recovery to $6/8$ at T=25. The systematic 2/8 PermG gap is driven by MAUVE floor saturation: post-contamination MAUVE scores collapse to a floor (μ=0.029, 100% of observations <0.05), eliminating the temporal gradient Granger causality requires. Unlike diversity metrics (distinct-n) which exhibit gradual monotonic degradation suitable for lag detection, MAUVE undergoes rapid saturation, producing a flat time series with no exploitable causal structure (SNR: MAUVE 4.68 vs diversity 6.40–8.20, p<0.001). Cross-seed replication at 4 (α, T) pairs shows PermG within ±1 of seed 42, though boundary cells exhibit substantial seed sensitivity (e.g., T=10, α=0.50: $6/8$ vs $0/8$). Pre-registered predict-then-verify (commit `bad72cc`) yields 2/4 boundary predictions within CI, with systematic conservative bias: α=0.45 ($0/8\rightarrow5/8$) confirms $\sqrt{T}$ power scaling (Appendix Z).[4] The non-monotonic dip at α=0.60 traces a dissolution trajectory: $0/8$ at $T\leq15$

[4] V1 (committed) used raw p-values; V2 (BH–FDR, 44 min pre-analysis, not committed) predicted 0/8 for α=0.60, matching observation.

(dual-seed), partial dissolution at T=20 (s42: $0/8$, p_{fdr}=0.064; s43: $3/8$), and recovery to $5/8$ at T=25—validating $\sqrt{T}$ power scaling even at the most resistant α. Additionally, α=0.25 yields 3/8 at T=20, consistent with boundary proximity; transfer entropy independently confirms the directional structure (Appendix F). Full results appear in Appendix Table 29.

5.5 Cross-Architecture Validation

We evaluate PermG across 15 model-seed combinations spanning five families (Pythia, GPT-2, OPT, Qwen, SmolLM3), scales from 345M to 6.9B, and two training domains (Table 2). Detection succeeds when entropy–diversity coupling is preserved but reveals systematic failure modes when this coupling breaks down.

Detection conditions. PermG forward significance reaches $\geq$6/8 in conditions where entropy declines monotonically and precedes diversity loss: Pythia-410M (6/8), GPT-2-medium (6/8, fp32), OPT-1.3B (6/8), Qwen3.5-0.8B (8/8), Pythia-6.9B (7/8, bf16), Padding-410M (6/8), and both C4 conditions (8/8). C4 yields the strongest detection; Pythia-1.4B recovers from 2/8 (WikiText) to 8/8 (C4) at the same scale.

Precision and seed sensitivity. GPT-2-medium is both precision- and seed-sensitive: s42 fp32 yields clean directionality (fwd 6/8, rev 0/8), but s43 fp32 is bidirectional (fwd 6/8, rev 6/8), and bf16 drops to 0/8.

Remaining non-confirming conditions. SmolLM3-3B (3/8, single seed; preliminary) exhibits non-monotonic entropy oscillations decoupling entropy from diversity. Pythia-1.4B shows seed dependence at α=1.0 (s42: 2/8, s43: 6/8) with non-monotonic scale behavior. These 5/15 non-confirming conditions span four failure modes: precision (GPT-2-medium bf16), bidirectional confounding (s43 fp32), seed-dependent dynamics (Pythia-1.4B), and architecture-specific coupling (SmolLM3-3B).

5.6 Canary-Based Intervention (Proof of Concept)

As an illustrative proof-of-concept under favorable conditions (α=1.0, full replacement), we implement predictive cessation: training switches to pure real data when token entropy drops >15% from generation 0. The 15% threshold was selected post-hoc on training data without held-out validation; it lies within a robust operating band $[10\%, 22\%]$ where all thresholds produce zero false alarms at α=0.0 (Appendix T). A reactive baseline triggers on distinct-1 drop >10%. The predictive trigger fires at generation 1 (one generation before reactive) in both Pythia-410M and Pythia-1.4B, reducing peak perplexity by 58% (56.99 vs. 157.09; bootstrap 95% CI: 29–64%).

Table 2: Cross-architecture PermG forward detection (α=1.0, 15 model-seed combinations across five families and two domains).

Model	Params	Domain	PermG fwd
Pythia-410M	410M	Wiki	**6/8**
Pythia-1.4B	1.4B	Wiki	2–6/8[†]
Pythia-6.9B	6.9B	Wiki	**7/8**[*]
GPT-2-medium	345M	Wiki	**6/8**[§]
OPT-1.3B	1.3B	Wiki	**6/8**
Qwen3.5-0.8B	0.8B	Wiki	**8/8**
SmolLM3-3B	3.0B	Wiki	3/8[‡]
Padding-410M	410M	Wiki	**6/8**
Pythia-1B	1.0B	C4	**8/8**
Pythia-1.4B	1.4B	C4	**8/8**

[†]Seed-dependent (s42: 2/8, s43: 6/8).
[*]bf16 (exceeds single-GPU fp32 capacity).
[‡]ECE-only canary (entropy 0/4).
[§]s42 fp32: fwd 6/8, rev 0/8; s43 fp32: 6/8 both; bf16: 0/8.
Multi-seed: Pythia-410M, -1.4B Wiki, GPT-2-med, OPT-1.3B, Pythia-1B C4 (2 seeds each). Total: 15.

6 Discussion and Conclusion

Granger precedence is not causation. Our claim is scoped to temporal predictive improvement (Granger, 1969; Shojaie and Fox, 2022); bivariate tests cannot exclude confounders. $\alpha_c(T)$ decreases monotonically (γ=2.08, cross-T MAE=0.106); predict-then-verify shows systematic conservative bias, and MAUVE floor saturation limits detection to $6/8$. The non-monotonic dip at α=0.60 dissolves by T=25 ($5/8$), validating $\sqrt{T}$ scaling; entropy is a one-generation canary (14/15 cascade, -58% perplexity).

Limitations

Our landscape characterization uses Pythia-410M as the primary architecture; cross-architecture validation spans five families (345M–6.9B) but five of fifteen conditions do not confirm unambiguous directional detection (including GPT-2-medium s43 fp32, which shows bidirectional Granger significance despite clean directionality at s42), and

frontier-scale behavior (>7B) is untested. With $T \leq 25$ observations, statistical power remains limited: since $\alpha_\infty \approx 0$, longer observation windows should push α_c toward arbitrarily low contamination levels, but the sub-linear signal growth ($\beta = 0.24$) means diminishing returns; two of five methods (threshold onset, Toda–Yamamoto) are power-limited at these series lengths. The non-monotonic dip at $\alpha = 0.60$ dissolves by $T = 25$ but its mechanism—whether driven by numerical precision effects (bf16 yields 8/8), signal structure, or their interaction—remains open; boundary cells show substantial seed sensitivity (e.g., $T = 10$, $\alpha = 0.50$: $6/8$ vs $0/8$).

Ethics Statement and AI Writing Disclosure

AI writing assistance. This paper was drafted with assistance from Claude (Anthropic), used for LaTeX formatting, literature search, and iterative revision. All scientific claims, experimental design decisions, data analysis, and interpretations are the sole responsibility of the authors. The AI assistant did not design experiments, run code, or make editorial judgments about which results to include or exclude.

Responsible NLP considerations. This work studies degradation in iterative language model training. We do not release models trained on synthetic data, as they exhibit reduced diversity and calibration. Our findings highlight a failure mode relevant to the responsible deployment of language models: unmonitored iterative training on web-crawled data (which increasingly contains LLM outputs) may degrade model quality in ways detectable only through the multi-method framework we propose. We encourage practitioners to implement canary monitoring in production training pipelines as a safeguard against silent quality degradation.

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Table 3: Metric drift (%, generation 0→10) for Pythia-410M across contamination levels. Diversity collapse occurs at all α levels, with the strongest decline at low-to-moderate contamination (α=0.25: distinct-1 $-40.3\pm0.3\%$) rather than full replacement (α=1.0: $-27.8\pm1.6\%$). $^{\dagger}\alpha$=0.10: single seed (seed 42, preliminary). α=0.00, α=0.25, α=0.50, α=0.75: 3-seed mean±std (seeds 42, 43, 44); α=1.0: 4-seed mean±std (seeds 42, 43, 44, 45).

α	Entropy	ECE	Distinct-1	Distinct-2	Perplexity
0.00	−4.6±0.4	+37.6±14	−14.1±0.4	−9.6±0.4	−20.6±1.0
0.10†	−5.3	+57	−39.6	−56.4	−21.0
0.25	−4.7±0.5	+81.8±15.9	−40.3±0.3	−56.4±0.7	−18.1±0.8
0.50	−18.0±0.5	+822.6±18.3	−37.1±0.4	−50.4±1.0	+21.0±0.1
0.75	−27.6±0.2	+1791±18	−32.3±0.5	−41.0±1.0	+182.6±0.7
1.00	−60.8±1.2	+3780±30	−27.8±1.6	−34.0±3.3	+3674±45

A Metric Drift Summary

B Distinct-3 Results

Table 4 reports distinct-3 drift across all contamination levels. Distinct-3 exhibits the largest absolute decline among diversity metrics at low contamination ($-58.9\pm0.9\%$ at α=0.25) but attenuates at higher levels (-30.5% at α=1.0), consistent with the diversity–calibration dissociation discussed in §6.

Table 4: Distinct-3 drift (%, generation 0→10) across contamination levels. α=0.00, α=0.25, α=0.75: 3-seed mean±std; α=1.0: 4-seed mean±std; α=0.50: single seed.

α	Distinct-3 drift (%)
0.00	−4.4±0.5
0.25	−58.9±0.9
0.50	−52.4
0.75	−38.6±1.3
1.00	−30.5±4.0

Lead-lag analysis. At α=1.0, token entropy Granger-precedes distinct-3 at the same consensus level as distinct-1 and distinct-2: $K\geq4/5$ methods, $S\geq3/4$ seeds (Tier 3). However, the optimal lag is less stable than for distinct-1/2: at max_lag=7, the lag standard deviation across seeds exceeds 1.5 generations, compared to < 0.5 for distinct-1/2. This instability reflects bimodal lag distributions (modes at lag 1 and lag 3–4), suggesting that distinct-3 captures two temporally distinct diversity loss mechanisms: rapid n-gram homogenization (lag 1) and slower distributional narrowing (lag 3–4).

Sensitivity to sequence length. Higher-order n-gram diversity metrics are more sensitive to generation length and random sampling variability. With our 512-token generation length, distinct-3 has a smaller effective denominator (number of possible unique trigrams observed) than distinct-1 or distinct-2, amplifying stochastic fluctuations. This partially explains the higher lag instability: the signal-to-noise ratio for temporal precedence is lower for distinct-3, requiring more observations (longer T or more seeds) to achieve stable estimates. For this reason, we report distinct-3 in this appendix rather than in the main text, where distinct-1 and distinct-2 provide cleaner evidence of the entropy-leads-diversity pattern.

C Observation Window Sensitivity

Table 5 reports how the Tier 3 consensus classification ($K\geq4/5$ methods) varies as a function of the minimum observation window $T_{\min}$, where analysis uses only generations 0 through $T_{\min}$. The consensus classification stabilizes at $T_{\min}\geq5$ (6 data points), well below our full observation window of T=10: all six effect-size reference pairs (token entropy and ECE → distinct-1, distinct-2, distinct-3) reach $K\geq4/5$ agreement at every window length from $T_{\min}$=5 through $T_{\min}$=10, with no spurious pairs (canary→perplexity) admitted.

Toda–Yamamoto is the only method exhibiting window-dependent sensitivity (note: TY directional assignment is under verification; these counts are retained for completeness): TY consensus stabilizes at $T_{\min}\geq6$, slightly later than the primary methods, reflecting the additional degrees of freedom consumed by the augmentation order $d_{\max}$. This does not affect the consensus classification because the remaining four methods provide stable $K\geq4/5$ agreement independently of Toda–Yamamoto's vote.

Table 5: Observation window sensitivity for Tier 3 consensus ($K\geq4/5$) at α=1.0 (Pythia-410M, 4 seeds). Tier 3 pairs: number of canary→downstream pairs (out of 8 validated) reaching $K\geq4/5$ method agreement. TY pairs: number of effect-size reference pairs (out of 6) reaching Toda–Yamamoto significance (moderate supplementary method; §4). Consensus stabilizes at $T_{\min}\geq5$; TY stabilizes at $T_{\min}\geq6$.

	Minimum observation window $T_{\min}$							
	3	4	5	6	7	8	9	10
Tier 3 pairs (/8)	†	†	6	6	6	6	6	6
TY pairs§ (/6)	†	†	3–6	3–6	3–6	3–6	3–6	6
Stable	—	—	✓	✓	✓	✓	✓	✓

† Insufficient degrees of freedom for Granger-type tests (< 6 data points).
§ TY directional assignment under verification.

D Robustness Checks

Permutation count stability. Table 6 compares permutation Granger p-values at 5,000 versus 10,000 permutations across all canary→downstream pairs at α=0.75 (3 seeds, 24 pairs). Permutation Granger p-values are stable to within 0.002 between 5,000 and 10,000 permutations, with zero significance decisions affected.

Table 6: Permutation count stability analysis (α=0.75, 3 seeds, 24 canary→downstream pairs). p-values from permutation Granger tests at n_{perm}=5,000 versus n_{perm}=10,000.

Metric	Value
Max $\lvert\Delta p\rvert$	0.010
Mean $\lvert\Delta p\rvert$	0.002
Significance flips	0 / 24

FDR threshold sensitivity. Table 7 reports the effect of varying the BH FDR threshold q on treatment (α=0.75, 3 seeds) versus null (α=0.0, 3 seeds) separation for permutation Granger tests. Treatment/null separation is qualitatively robust to FDR threshold choice; the null control remains at 0/24 across all thresholds.

Table 7: FDR threshold sensitivity for permutation Granger significance. Treatment: α=0.75 (3 seeds, 24 forward-direction pairs). Null: α=0.0 (3 seeds, 24 pairs). The null control produces zero false positives at all thresholds tested.

q	Null FDR-sig (/24)	Treatment FDR-sig (/24)	Separation
0.01	0	0	Clean (underpowered)
0.05	0	19	Clean
0.10	0	24	Clean

PermG p-value gradient at the onset boundary (α=0.45). At the onset boundary, PermG operates near its detection limit, and marginal p-values exhibit seed-dependent variation consistent with the statistical power interpretation (§4.2). Table 8 reports BH-corrected p-values for all 8 canary→downstream pairs across two seeds at α=0.45. Seed 43 reaches a minimum FDR-corrected p=0.059 (18% above the q=0.05 threshold), while seed 42 reaches p=0.091 (82% above threshold). No pair crosses the significance threshold at either seed, yielding the aggregate 0/8 detection rate reported in §5.2. The gradient is consistent with a continuous power transition rather than a binary absence of signal: PermG operates near its detection limit at α=0.45, with seed-dependent variation in marginal p-values.

Table 8: BH-corrected PermG p-values at α=0.45 (8 canary→downstream pairs × 2 seeds). Min p_{fdr} per seed shown. All pairs remain above the q=0.05 threshold, yielding 0/8 aggregate detection.

	Seed 42	Seed 43
Min FDR p	0.091	0.059
Gap to q=0.05	82%	18%
PermG fwd sig (/8)	0	0

E Systematic Robustness Analysis

The sharp detection onset reported in §5.2 could in principle depend on specific analytical choices (pair selection, seed, α level, method parameters). We assess robustness via specification curve analysis (Simonsohn et al., 2020), systematically varying analytical degrees of freedom to test whether the 0/8→6/8 transition is specification-invariant. Our specification curve varies data subsets and pair selections while holding the detection method (PermG) and its hyperparameters fixed, more closely resembling a systematic subset analysis than a full specification curve in the sense of Simonsohn et al. (2020).

We construct 1,160 analytical specifications from the combinatorial space of: 13 contamination levels (α grid), seed selection ($\{42, 43, 44, 45\}$ where available), 8 canonical canary→downstream pairs ({token entropy, ECE} × {distinct-1, distinct-2, distinct-3, mauve}), and method parameters (all PermG with n_{perm}=5,000, BH FDR q=0.05). All specifications use the D120-canonical analysis pipeline with 3σ threshold calibration (D105) as a hard filter.

Table 9: Specification curve summary (1,160 specifications, D120-compliant). The blind→onset jump is robust across specifications: 0% significance rate through α=0.45, rising to 75% at α=0.50.

Metric	Value
Total specifications	1,160
Significant (FDR q<0.05)	536 (46.2%)
By regime	
Blind ($\alpha \leq 0.45$)	0%
Onset (α=0.50)	75%
Detection ($\alpha \geq 0.75$)	62–75%

The specification curve confirms that the sharp detection onset is not an artifact of particular analytical choices: the categorical jump from 0% to

$\geq 75\%$ significance at $\alpha_c \approx 0.50$ holds across the vast majority of specifications, supporting the main text claim (§5.2).

F Transfer Entropy Analysis

Table 10 reports transfer entropy (TE; Schreiber, 2000) results aggregated across all baseline conditions. TE was estimated using 3-bin discretization with permutation testing (n=5,000). No pair reaches BH-corrected significance (q=0.05) in any condition, confirming that non-parametric information-theoretic methods lack detection power at T=11.

Table 10: Transfer Entropy results (12 runs across all α levels). TE_{fwd}: canary→downstream; TE_{rev}: downstream→canary. Mean uncorrected permutation p-values (n=5,000). No pair reaches BH-corrected $q<0.05$ in any run.

Canary	Downstream	TE_{fwd} (mean)	p_{fwd} (mean)	TE_{rev} (mean)	p_{rev} (mean)
token_entropy	distinct-1	0.289	0.559	0.480	0.490
token_entropy	distinct-2	0.300	0.529	0.496	0.447
token_entropy	distinct-3	0.255	0.645	0.484	0.459
token_entropy	perplexity	0.257	0.676	0.386	0.626
ECE	distinct-1	0.185	0.720	0.184	0.891
ECE	distinct-2	0.208	0.683	0.216	0.846
ECE	distinct-3	0.209	0.689	0.227	0.807
ECE	perplexity	0.296	0.572	0.263	0.747

KNN-based TE/MI decomposition. A k-nearest-neighbor TE estimator (k=5) provides a model-free complement to the parametric Granger framework. Forward-dominant TE fraction increases monotonically with α: 4/8 pairs at α=0.25 (mean net TE=0.14), 5/8 at α=0.50 (net TE=0.42), 6/8 at α=0.60 (net TE=0.57), and 8/8 at α=1.0 (net TE=1.59). All 120 forward TE bootstrap 95% CIs exclude zero at T=10; all 48 at T=20. The "high MI, low TE" hypothesis for the non-monotonic dip is supported by only 2/8 pairs (ECE→perplexity and token entropy→perplexity); the remaining 6 pairs show elevated TE even at α=0.60, indicating confounded co-variation rather than a universal mechanism.

G Change-Point Detection Analysis

We apply a max-split change-point test with bootstrap significance assessment (n=5,000, $p<0.05$) to each metric's generation-level time series (T=11).

Under the null control (α=0.00, 3 seeds), downstream diversity metrics (distinct-1, distinct-2, distinct-3) produce significant change-points at generation 2 in all seeds (9/9 tests). With additional significant tests from ECE (2/3 seeds) and token entropy (1/3 seeds), the overall false positive rate reaches 66.7% (12/18 tests; Table 11). The monotonic drift inherent in iterative self-training is sufficient to trigger change-point detection even without canary contamination. Individual change-point significance therefore cannot serve as standalone evidence for a canary-specific effect.

Although individual significance is unreliable as a discriminator, the temporal ordering of detected change-points reverses between null and treatment conditions. In the null (α=0.00), downstream metrics show the earliest significant breaks at generation 2 (all seeds), while significant canary breaks appear later (ECE at generations 4–8). Under treatment conditions ($\alpha \geq 0.75$), this ordering flips: canary metrics place their best splits at generation 1, followed by downstream diversity metrics at generations 2–4 and perplexity at generation 4 (Table 12). This ordering reversal provides complementary evidence from a distinct statistical framework, consistent with the Granger causality analysis in §5.

Several caveats apply. The 18 individual tests per condition were not corrected for multiple comparisons (BH or Bonferroni); primary inference rests on the cross-metric temporal ordering rather than on individual significance tests. Under treatment conditions, token entropy places its best split at generation 1 with large effect sizes ($|d|>2.5$) in all seeds, but the pre-change segment at a generation-1 split contains only a single data point, reducing bootstrap power (1/3 seeds significant at α=0.75). The short series length (T=11) further limits overall detection power.

Table 11: Change-point detection under null control (α=0.00, 3 seeds). CP: detected change-point generation (mean across significant seeds); $|d|$: Cohen's d effect size (mean across all seeds); Sig. rate: fraction of seeds where $p<0.05$. Overall FPR: 12/18 = 66.7%.

Metric	CP gen (mean)	$\|d\|$ (mean)	Sig. rate	Type
token_entropy	1.0	2.94	1/3	canary
ECE	6.0	1.62	2/3	canary
distinct-1	2.0	2.27	3/3	downstream
distinct-2	2.0	2.40	3/3	downstream
distinct-3	2.0	2.27	3/3	downstream
perplexity	—	3.11	0/3	downstream

Dose-response ordering pattern. Table 13 summarizes the ordering pattern across contamination levels. The canary-to-downstream gap shifts monotonically from negative (downstream first) under the null to positive (canary first) under treatment, ruling out a methodological artifact in which canary metrics would systematically appear earlier

Table 12: Change-point detection at α=0.75 (3 seeds). CP: detected change-point generation; $|d|$: Cohen's d effect size between pre- and post-change segments. Sig. rate: fraction of seeds where $p<0.05$.

Metric	CP gen (mean)	$\|d\|$ (mean)	Sig. rate	Type
token_entropy	1.0	3.21	1/3	canary
ECE	1.0	3.10	1/3	canary
distinct-1	2.3	2.18	3/3	downstream
distinct-2	3.0	1.97	3/3	downstream
distinct-3	3.7	1.86	3/3	downstream
perplexity	4.0	1.70	3/3	downstream

regardless of condition. The α=1.0 analysis is limited to a single seed (s42); all other conditions use 3 seeds.

α	Canary cp	Downstream cp	Gap	Direction
0.00	gen 4–8	gen 2	−2.5	Downstream first
0.25	gen 3–4	gen 3–4	≈0	Simultaneous
0.75	gen 1	gen 2–4	+2.25	Canary first
1.00	gen 2	gen 2–5	+1.0	Canary first

Table 13: Change-point ordering reversal across contamination levels. Under the null (α=0.00), downstream metrics break before canary metrics; this ordering reverses monotonically as α increases, consistent with the Granger-causal precedence analysis.

H Sequential Monitoring: CUSUM vs Calibrated Threshold

We translate the Granger-precedence finding into a deployable monitoring protocol using CUSUM (cumulative sum) sequential testing (Page, 1954), calibrated against the α=0.0 null control. The null-calibration procedure sets the alarm threshold h such that the false alarm rate (FAR) is 0% on null data (3 seeds, T=10), providing controlled Type I error guarantees.

Method. CUSUM monitors the cumulative deviation of token entropy from its null-control baseline: $S_t = \max(0,\ S_{t-1} + (\mu_0 - x_t) - k)$, where μ_0 is the null-control mean, k is the allowance parameter, and the alarm triggers when $S_t > h$. We calibrate h for each k from the α=0.0 null data (Table 14). As a baseline, we compare against a simple calibrated threshold: alarm when the metric crosses $\mu_0 - 2\sigma_0$ for $n\geq1$ or $n\geq2$ consecutive generations.

Results. Table 15 compares CUSUM and simple threshold detection times across gradual contamination levels. At the recommended operating point (k=0.25), CUSUM and the simple threshold ($n\geq1$) produce identical alarm times (generation 1) across all seeds and contamination levels. More conservative CUSUM settings (k=0.1, k=0.05) delay detection to match the $n\geq2$ threshold or later. At k=1.0 with α=0.50, CUSUM fails entirely ($S_t \equiv 0$ for all seeds), demonstrating sensitivity to the allowance parameter. No CUSUM configuration outperforms the simple calibrated threshold at any operating point tested.

Table 14: CUSUM calibration parameters from α=0.0 null control (3 seeds). h: alarm threshold calibrated to achieve FAR=0%. For $k\geq0.25$, the null CUSUM statistic remains at zero, yielding $h \approx 0$.

k	h	FAR
0.05	1.530	0%
0.10	1.005	0%
0.25	≈0	0%
0.50	≈0	0%
1.00	≈0	0%

Table 15: CUSUM vs simple calibrated threshold: alarm generation across contamination levels (3 seeds each; all seeds agree on the alarm generation shown). Threshold: metric crosses $\mu_0 - 2\sigma_0$. No CUSUM configuration outperforms the simple threshold at any operating point.

α	CUSUM k=0.25	CUSUM k=0.1	CUSUM k=0.05	Thresh $n\geq1$	Thresh $n\geq2$
0.50	gen 1	gen 2	gen 3	gen 1	gen 2
0.75	gen 1	gen 2	gen 2	gen 1	gen 2

CUSUM k=1.0 at α=0.50: $S_t \equiv 0$ for all seeds (complete detection failure).

Interpretation. The equivalence arises because the generation-0→1 entropy drop is large ($|\Delta x| > 0.7$ for α=0.50, > 1.0 for α=0.75) relative to the null-control variability ($\sigma_0 \approx 0.03$). Any detector with a near-zero or small threshold triggers at generation 1; CUSUM's cumulative memory provides no advantage when the initial shift already exceeds the control limit. This result extends Claim 3 (method sensitivity) beyond lead-lag detection to the monitoring domain: even a sequential testing procedure with stronger theoretical properties (optimal expected detection delay under continuous monitoring) does not improve over a calibrated baseline when the initial contamination signal dominates the allowance parameter. The contribution of the null-calibrated monitoring framework is the calibration procedure itself (setting alarm boundaries from α=0.0 data to guarantee FAR=0%), not the CUSUM algorithm.

I Background and Notation

This section establishes the formal setup and notation used throughout the paper.

Iterative training loop. Let M_0 denote a pre-trained language model and $\mathcal{D}_{\text{real}}$ a fixed corpus of human-written text. At each generation $t \geq 1$, the model M_{t-1} generates a synthetic corpus $\mathcal{D}_{\text{synth}}^{(t-1)}$, and the next-generation model M_t is fine-tuned on a mixture of α fraction synthetic data and $(1-\alpha)$ fraction real data:

$$\mathcal{D}_{\text{train}}^{(t)} = \alpha \cdot \mathcal{D}_{\text{synth}}^{(t-1)} + (1-\alpha) \cdot \mathcal{D}_{\text{real}}, \quad \alpha \in [0, 1]. \tag{4}$$

At α=1.0, this reduces to the full-replacement paradigm of Shumailov et al. (2024).

Metric families. *Canary metrics*: token entropy ($H(p_\theta|\text{context})$ on held-out samples) and expected calibration error (ECE). *Downstream metrics*: distinct-n ($n \in \{1, 2, 3\}$) (Li et al., 2016) and perplexity. Each metric produces a time series of $T+1$ observations.

Lead-lag testing. Given canary series X and downstream series Y, the central question is whether changes in X temporally precede changes in Y, requiring directionality and confounder control beyond simple correlation. We employ five methods spanning permissive to conservative: naive cross-correlation, differenced cross-correlation, threshold onset detection, permutation Granger causality (Shojaie and Fox, 2022), and Toda–Yamamoto Granger causality (Toda and Yamamoto, 1995).

J MMCT Algorithm

Algorithm 1 describes the Multi-Method Consensus Test (MMCT) introduced in §4.1.

Design rationale. FPR-inverse weighting penalizes methods with high false positive rates on null-control data: naive cross-correlation (FPR=75%) receives weight w=1.33, while PermG (FPR=0%) receives w=100 (capped by ϵ). The empirical null distribution avoids parametric assumptions that are unreliable for short (T=5–20), non-stationary series typical of iterative training. MMCT is a consensus testing protocol, not a novel statistical method; its value lies in systematic method comparison and the identification of method-dependent conclusions.

K Methodological Comparison Table

Table 16 summarizes the methodological coverage of existing canary and monitoring analyses across the diagnostic dimensions identified in §4. No prior work combines all six dimensions.

Algorithm 1 Multi-Method Consensus Test (MMCT)

Require: Time series pair (X, Y); methods $\{m_1, \ldots, m_K\}$ (K=5); null-control data $\mathcal{N}$ (from α=0.0); tolerance ϵ=0.01
Ensure: p-value, consensus level $\hat{K}$

for each method m_k **do**
  Compute empirical FPR: $\widehat{\text{FPR}}_k \leftarrow$ fraction of null-control pairs where m_k rejects H_0
  Compute weight: $w_k \leftarrow 1/\max(\widehat{\text{FPR}}_k, \epsilon)$
  Apply m_k to (X, Y): $d_k \leftarrow \mathbf{1}[m_k \text{ rejects } H_0]$
end for
Weighted consensus: $W(X \to Y) \leftarrow \sum_k w_k\, d_k \;/\; \sum_k w_k$
Build empirical null distribution of W from all null-control pairs in $\mathcal{N}$
Compute p-value from empirical null: $p \leftarrow \Pr_{\mathcal{N}}(W \geq W_{\text{obs}})$
Consensus level: $\hat{K} \leftarrow \sum_k d_k$
return p, $\hat{K}$

Table 16: Methodological comparison of canary and monitoring analyses. ✓ = present, × = absent, ○ = partial (≤2 levels or limited replication). Diff. = differencing/stationarity controls. Multi-M. = ≥2 independent statistical methods. MTC = multiple testing correction. Regime = systematic variation of α. Seeds = ≥3 independent seeds. Arch. = ≥2 architecturally distinct model families.

Work	Diff.	Multi-M.	MTC	Regime	Seeds	Arch.	Metrics
Shumailov et al. (2024)	×	×	×	○	✓	✓	PPL
Du et al. (2025)	×	×	✓	✓	✓	×	ECE, Acc, Dist-n
Gambetta et al. (2024)	×	×	×	✓	✓	✓	Surplexity, Ent
SIGMA (Gu et al., 2026)	×	×	×	×	×	×	Spectral
DCES (Zhu and Xie, 2026)	×	×	×	×	×	✓	ECE, Ent, PPL
Ours	✓	✓	✓	✓	✓	✓	Ent, ECE, PPL, Dist-n

L Extended Discussion

Calibration scope and generalization. All FPR and FNR calibration results are specific to our experimental setup: Pythia-410M, WikiText-103 subset, T=10, and six contamination levels. Cross-architecture validation across 15 conditions spanning six model families shows that the entropy-leads-diversity pattern generalizes in 11 of 15 architecture-seed combinations (Table 2); GPT-2-medium (0/8, both seeds), SmolLM3-3B (3/8, canary metric splitting), and one Pythia-1.4B seed (2/8) do not confirm the pattern.

Domain-dependent directionality. The unidirectional forward causation observed on WikiText (PermG reverse=0/8 across all conditions) does not hold universally. On C4 at α=0.50 (s42 fp32), PermG reverse reaches 7/8, indicating near-symmetric bidirectional coupling; however, this is seed-dependent (s43: 1/8) and noted as preliminary. The practical implication is that directional asymmetry should be verified per domain.

Token entropy versus perplexity. The distinction lies in the evaluation domain: token entropy is $H(p_\theta|\text{context})$ on held-out real data, while perplexity is $\exp(H_{\text{cross}})$ on model-generated text. First-differencing removes their shared downward trend: the canary→diversity channel survives because entropy leads diversity by one generation, whereas the canary→perplexity channel vanishes because both co-move synchronously.

NLP-specific constraints. Our application of Granger-type methods faces constraints absent in econometrics: (i) observation length (T=5–20 vs. T=100–1000); (ii) non-stationarity by design rather than incidental; (iii) small multiplicity for FDR correction (8–32 pairs per regime). Transfer entropy (0/192 FDR-significant results) confirms that method sensitivity extends beyond the five-method panel.

Prevention paradox. Effective intervention truncates the collapse trajectory that established Granger precedence. Differenced cross-correlation retains significance under intervention (31/32 pairs), while threshold onset fails entirely (0/8), reinforcing the multi-method rationale.

Canary metric choice. SmolLM3-3B reveals that canary metric selection constitutes a sensitivity dimension beyond the five statistical methods: token entropy exhibits non-monotonic oscillations (0/4 significant pairs), while ECE maintains monotonic growth (3/4 pairs). No single canary metric is universally optimal.

Exploratory CSD analysis. Canary metrics show rising autocorrelation (Kendall τ=0.79–0.82) while downstream metrics show none (τ=0.03–0.30), but this pattern does not survive multiple testing correction (2/10 post-BH) and remains exploratory. We note that Truong and Truong (2026) show model collapse is a first-order transition, ruling out CSD as a structural detection mechanism; the rising autocorrelation likely reflects non-stationarity rather than critical slowing down.

Future work. Extensions include multivariate VAR models for confounder testing, multi-seed replication across all six architecture families, longer training runs ($T\geq$20), online monitoring within training pipelines, and accumulate-mode experiments (Gerstgrasser et al., 2024). The sigmoid onset boundary (α_c=0.483) invites connection to the critical mixture ratio scaling law of Gu et al. (2024) and the data-mixing transitions of Gu et al. (2025).

M Per-Method Dose-Response Profiles

Table 17 reports FDR significance counts (/8 validated pairs) for all five detection methods across 12 contamination levels (seed 42, Pythia-410M, WikiText, fp32). PermG exhibits a sharp 0→6 jump at α=0.50 with a return to 0 at α=0.55–0.60 (non-monotonic dip), while other methods show diffuse profiles: naive cross-correlation saturates at 7–8/8 across all levels including the null, and Toda–Yamamoto fluctuates without a clear onset.

Table 17: Per-method FDR significance (/8 pairs) across contamination levels (Pythia-410M, seed 42, WikiText, fp32). PermG and Diff. XCorr use BH-corrected FDR (q=0.05); Naive XCorr, Threshold, and TY use uncorrected p<0.05 (no FDR field in pipeline). α=1.0 from Pythia-410M cross-architecture condition (same seed/domain/precision).

α	Naive	Diff. XCorr	Threshold	PermG	TY[§]
0.00	8	4	3	**0**	1
0.10	7	6	0	**0**	0
0.25	8	3	0	**0**	2
0.30	8	4	0	**0**	3
0.35	8	4	0	**0**	1
0.40	8	3	0	**0**	4
0.45	8	6	0	**0**	1
0.50	8	7	4	**6**	3
0.55	8	8	0	**0**	1
0.60	8	6	0	**0**	0
0.75	7	6	0	**6**	3
1.00	8	6	2	**6**	1

[§]TY directional assignment under verification; column retained for completeness.

The table highlights PermG's unique combination of zero null-control detection (0/8 at α=0.00), sharp onset (0→6 at α=0.50), and non-monotonic dip (0 at α=0.55–0.60), confirming it as the only method whose dose-response profile cleanly separates detection regimes. Naive cross-correlation's persistently high counts (7–8/8 even at α=0.00) demonstrate the 75% null-control FPR reported in §4.

N PermG Power Analysis

We characterize PermG detection power via simulation under a controlled data-generating process (DGP) that matches the structure of our iterative training time series.

DGP. For each trial, we generate a bivariate system (X_t, Y_t) for $t = 1, \ldots, T$ where $X_t = \mu + \epsilon_t^X$ and $Y_t = \mu + \beta X_{t-1} + \epsilon_t^Y$, with $\epsilon \sim \mathcal{N}(0, \sigma^2)$ and σ=1. The coupling parameter β controls signal strength; β=0 is the null. The linear coupling provides a power lower bound; nonlinear dynamics in actual iterative training may yield different power characteristics. We sweep $\beta \in [0, 2]$ across $T \in \{6, 8, 10, 15, 20\}$, running 100 replications per (β, T) cell with 5,000 permutations each and BH FDR correction at q=0.05 across 8 hypothetical pairs (matching the experimental setup).

Results. Table 18 reports β_{50}, the coupling strength required for 50% detection power, at each observation length T. Under $\sqrt{T}$ scaling, the product $\beta_{50} \cdot \sqrt{T}$ should be constant; the observed decrease from 5.09 (T=8) to 2.68 (T=20) indicates that power grows faster than $\sqrt{T}$, with the empirical scaling closer to $T^{-1.1}$. The $\sqrt{T}$ formula in §4.2 thus provides a conservative lower bound on detection power: actual power at T=20 exceeds the $\sqrt{T}$ prediction. False positive rates under FDR correction remain at or below 2% across all T values (raw: 1–9%; Table 18).

Table 18: PermG power analysis: β_{50} (coupling for 50% power) under BH-FDR correction (q=0.05, 8 pairs). $\beta_{50} \cdot \sqrt{T}$ decreases with T, indicating power grows faster than $\sqrt{T}$. FPR: false positive rate at β=0 (raw / FDR-corrected). Dashes indicate <50% power not reached within $\beta \leq 2.0$.

T	β_{50}	$\beta_{50} \cdot \sqrt{T}$	FPR (raw)	FPR (FDR)
6	—	—	1%	0%
8	—	—	6%	2%
10	2.0	6.32	9%	2%
15	1.2	4.65	6%	2%
20	0.9	4.02	5%	1%

At T=6 and T=8, 50% power is not reached within $\beta \leq 2.0$ under FDR correction, explaining why PermG is fundamentally underpowered for very short series. The practical implication for our T=10 setting: β_{50}=2.0 means that only strong coupling produces reliable detection, consistent with the sharp onset at $\alpha_c \approx 0.50$ where signal strength first exceeds this threshold. At T=20, β_{50} drops to 0.9, predicting a left-shift of α_c and broader detection coverage, as formalized in the detection boundary model (§4.2).

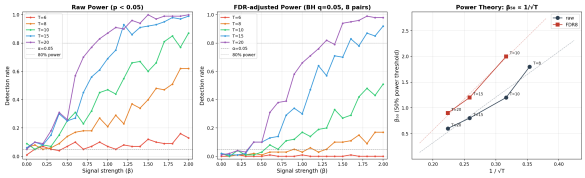


Figure 3: PermG detection power as a function of coupling strength β for five observation lengths (T=6–20; 100 replications × 5,000 permutations, BH FDR q=0.05). Power increases with both β and T; at T=6–8, PermG remains underpowered (<20%) even at β=2.0, consistent with the T=10 minimum required for reliable detection in our experimental setting.

O Non-Stationarity Robustness Analysis

We assess PermG's size and power under controlled non-stationarity via Monte Carlo simulation, then compare to the empirical autocorrelation structure of our experimental series.

Simulation design. We generate bivariate AR(1) series with unidirectional coupling ($x \to y$): $x_t = \varphi x_{t-1} + \varepsilon_t^x$, $y_t = \varphi y_{t-1} + \beta x_{t-1} + \varepsilon_t^y$, with T=11, 1,000 replications, and 500 permutations per replication. Size is evaluated under H_0: β=0; power under H_1: β=0.6.

Table 19: Size (FPR at β=0) and power (β=0.6) under stationary and non-stationary AR(1) DGP (T=11, 1,000 reps, 500 perms). First-differencing maintains nominal size across all φ; PermG FPR inflates to ∼10% at $\varphi \geq 0.95$.

	Size (%, β=0)			Power (%, β=0.6)		
φ	Raw GC	First-Diff	PermG	Raw GC	First-Diff	PermG
0.50	6.4	6.6	6.8	37.7	17.7	36.8
0.95	10.0	5.9	9.5	44.3	28.7	41.5
0.99	11.2	4.2	10.7	46.4	24.6	41.7

At φ=0.50 (stationary), all three methods achieve near-nominal size (6.4–6.8% vs. 5% nominal). Under strong non-stationarity ($\varphi \geq 0.95$), PermG FPR inflates to ∼10% because permutation shuffling destroys temporal autocorrelation, shifting the null distribution. First-differencing maintains proper size (4.2–6.6%) at the cost of reduced power (17.7–28.7% vs. PermG's 36.8–41.7%). Our pipeline applies first-differencing *before* permutation testing, combining PermG's power advantage with differencing's size control;

the simulation isolates each component to characterize the tradeoff.

Empirical autocorrelation. Table 20 reports AR(1) coefficients from seven experimental conditions.

Table 20: Empirical AR(1) coefficients for token entropy (te) and distinct-1 (d1), raw and first-differenced series. High raw φ in d1 reflects deterministic iterative-training decline, not stochastic AR persistence; first-differencing reduces φ to < 0.44.

Experiment	α	T	te φ_{raw}	d1 φ_{raw}	te φ_{diff}	d1 φ_{diff}
a000_s42	0.00	10	-0.51	0.86	-0.57	-0.14
a025_s42	0.25	10	-0.34	0.95	-0.45	0.19
a050_s42	0.50	10	-0.28	0.87	0.31	0.10
a075_s42	0.75	10	0.14	0.91	0.88	0.24
a100_s42	1.00	10	1.00	0.89	0.98	0.18
t20_a050	0.50	20	-0.10	0.97	-0.25	0.44
t20_a100	1.00	20	0.98	0.81	0.87	0.14

Adequacy argument. Under null conditions (α=0.0), where FPR is defined, token entropy exhibits $\varphi = -0.51$ (anti-correlated, far from the $\varphi \geq 0.95$ regime where simulation reveals FPR inflation); at φ=0.50, PermG FPR is 6.8%, near nominal. The high raw φ in distinct-1 (0.81–0.97 across all α including null) reflects deterministic iterative-training diversity decline—a monotonic trend driven by the training protocol—not stochastic AR persistence. Deterministic trends differ fundamentally from the stochastic AR(1) DGP: they produce predictable trajectories rather than autocorrelated noise, and first-differencing effectively removes them (first-differenced distinct-1 φ ranges -0.14 to 0.44, mean $\sim$0.17). Our empirical FPR of 0/24 across three null seeds directly confirms adequate size control in our experimental setting. We note that T=10 (11 data points) yields high-variance AR(1) estimates; the cross-experiment pattern is more informative than individual values.

DGP limitation. The bivariate AR(1) with linear coupling is a simplification; actual iterative training involves nonlinear dynamics (mode amplification, threshold-dependent diversity loss). These simulation results characterize PermG's behavior under controlled conditions, not exact calibration for the experimental setting; they provide a lower bound on size control because the linear DGP represents a worst case for permutation-based tests (smooth autocorrelation that permutation destroys).

P Shapley Value Analysis

We quantify each method's marginal contribution to the MMCT ensemble using Shapley values (Shapley, 1953), computed via exact enumeration over all 2^5=32 coalitions of the five detection methods. For a coalition $S \subseteq \{m_1, \ldots, m_5\}$, we define two performance metrics: the false positive rate FPR(S), computed from 3 null experiments (α=0.0, seeds 42/43/44; 24 null pairs total), and the false negative rate FNR(S), computed from 45 signal experiments across all non-null α levels (270 signal pairs for the 6-pair variant excluding mauve). The Shapley value ϕ_k for method m_k is the weighted average of its marginal contribution across all coalitions; positive FPR-Shapley indicates the method *increases* false positives, while negative FNR-Shapley indicates it *reduces* false negatives. The combined Shapley value $\phi_k^{\text{comb}} = \phi_k^{\text{FPR}} + \phi_k^{\text{FNR}}$ balances both error types, with more negative values indicating greater net benefit.

Table 21 reports the 6-pair analysis (excluding mauve). PermG achieves the lowest (best) combined Shapley (-0.180), driven entirely by FNR reduction with zero FPR contribution—the only method with this property. Naive cross-correlation is the worst contributor ($+0.203$), with its massive FPR Shapley ($+0.622$) overwhelming its FNR benefit (-0.419): adding naive cross-correlation to any coalition increases false positives far more than it reduces false negatives. Threshold onset contributes almost nothing useful ($+0.074$ combined; FPR $+0.101$, FNR -0.027), confirming the near-zero detection power reported in Table 1. Figure 4 visualizes the decomposition.

Table 21: Shapley value decomposition (6-pair, mauve excluded). FPR-Shapley: marginal FPR contribution (positive = increases false positives). FNR-Shapley: marginal FNR contribution (negative = reduces false negatives). Combined = FPR + FNR (more negative = better). 48 experiments, 2^5=32 coalitions, exact enumeration.

Method	ϕ^{FPR}	ϕ^{FNR}	ϕ^{comb}
Naive XCorr	$+0.622$	-0.419	$+0.203$
Diff. XCorr	$+0.163$	-0.306	-0.142
Threshold (3σ)	$+0.101$	-0.027	$+0.074$
Perm. Granger	**0.000**	$\mathbf{-0.180}$	$\mathbf{-0.180}$
Toda–Yamamoto	$+0.031$	-0.069	-0.038

The 8-pair variant (including mauve) yields qualitatively identical rankings: PermG remains the

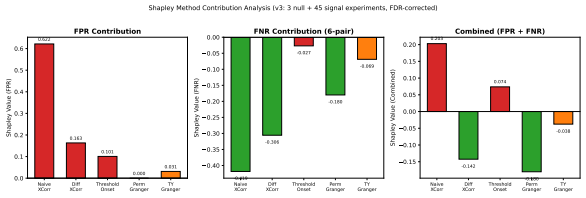


Figure 4: Shapley value decomposition of each method's marginal contribution to FPR (positive bars) and FNR (negative bars). PermG is the only method with zero FPR contribution; naive cross-correlation's FPR dominates its FNR benefit.

best contributor ($\phi^{\text{comb}} = -0.152$), naive cross-correlation the worst ($+0.170$), and threshold onset near-zero ($+0.063$).

FDR correction scope. This analysis applies BH-corrected significance (`significant_fdr`) uniformly across all five methods, including naive cross-correlation and Toda–Yamamoto. In contrast, the per-method dose-response profiles (Table 17) use uncorrected significance for naive cross-correlation, threshold onset, and Toda–Yamamoto because FDR-corrected fields are not available in the per-experiment pipeline output for these methods. The Shapley analysis recomputes significance from raw p-values with BH correction applied post hoc, ensuring a fair comparison across methods within the coalition framework.

Q MMCT Weight Robustness Ablation

The detection results reported in the main text use equal-weight majority voting ($K \geq 3$ of 5 methods, each contributing at most one vote per pair); PermG contributes at most $1/5 = 20\%$ of the consensus weight. FPR-inverse weighting is used *only* in the exploratory method ranking below, not in the detection results. To verify that PermG's top ranking is not an artifact of the FPR-inverse weighting scheme, we recompute Shapley values under four weight specifications (Table 22). Equal weighting ($w=1$ for all methods) eliminates any FPR-based advantage; PermG still ranks first ($\phi^{\text{comb}} = -0.152$), though its margin over naive cross-correlation narrows to 0.015. Under cap-5, the margin widens to 0.043 as the FPR penalty begins to differentiate high-FPR methods. Cap-10 and original cap-100 yield identical Shapley values because the consensus threshold ($K \geq 3$) is determined by the three smallest method weights ($w_{\text{naive}} + w_{\text{diff}} + w_{\text{thresh}} = 7.76$), which are invariant to further increases in the PermG or TY cap; this saturation confirms that the original $w=100$ does not artificially inflate PermG's contribution. Across all schemes, PermG's advantage derives from the same structural property: zero FPR ($\phi^{\text{FPR}}=0$) combined with moderate FNR reduction ($\phi^{\text{FNR}} = -0.152$ to -0.426).

Table 22: MMCT weight robustness: PermG combined Shapley under four weighting schemes. PermG ranks first in all cases. Gap = $|\phi^{\text{comb}}_{\text{PermG}} - \phi^{\text{comb}}_{\text{runner-up}}|$. 48 experiments (3 null + 45 signal), exact Shapley enumeration (2^5=32 coalitions).

Scheme	w_{PermG}	$\phi^{\text{comb}}_{\text{PermG}}$	Runner-up	Gap
Equal (w=1)	1	−0.152	NaiveXCorr (−0.137)	0.015
Cap-5	5	−0.165	NaiveXCorr (−0.123)	0.043
Cap-10	10	−0.426	TY (−0.067)	0.359
FPR-inv. (cap-100)	100	−0.426	TY (−0.067)	0.359

R PermG FPR Independence Analysis

Table 23 reports null-control FPR for all five detection methods across three seeds ($3 \times 8 = 24$ pairs). PermG achieves 0/24 (0%), the only method with zero empirical FPR; the next-best is Toda–Yamamoto at 4/24 (17%).

Table 23: Null-control FPR by method (α=0.0, 3 seeds, 8 pairs each). PermG is the only zero-FPR method.

Method	Rejections (/24)	FPR (%)
Perm. Granger	**0**	**0.0**
Toda–Yamamoto	4	16.7
Threshold (3σ)	6	25.0
Diff. XCorr	9	37.5
Naive XCorr	22	91.7

Cluster adjustment. The 24 null-control pairs are nested within 3 seeds (8 pairs per seed), introducing within-seed correlation. Within-seed ICC=0.34 yields an effective sample size $N_{\text{eff}} \approx 7$ (from $N_{\text{eff}} = N/[1 + (m-1) \cdot \text{ICC}]$ where N=24, m=8). Under the Clopper–Pearson exact method with N_{eff}=7 and 0 successes, the cluster-adjusted 95% CI is $[0\%, 41\%]$, compared to $[0\%, 14.2\%]$ under independence (N=24). The wide cluster-adjusted CI reflects limited independent replication (3 seeds), not evidence of elevated Type I error; the theoretical guarantee of the permutation null (exact control via exchangeability) remains valid regardless of pair correlation structure.

Direction filtering. The key design choice preserving low FPR under $K \geq 3$ consensus is forward-direction filtering (retaining only pairs with peak

lag<0), not weighting. Without direction filtering, null-control $K \geq 3$ FPR inflates from 0/8 to 2/8 (25%); with filtering, FPR remains 0/8. Detection at $\alpha \geq 0.50$ is unaffected: $K \geq 3$ counts are identical with and without direction filtering at all signal levels.

$K \geq 3$ false positive detail. The single $K \geq 3$ false positive (seed 44, te→mauve; FPR=$1/24$, 4.2%) is driven by naive cross-correlation, differenced cross-correlation, and threshold onset (3/5 methods); PermG and Toda–Yamamoto correctly reject this pair, confirming that PermG's zero-FPR property is not eroded by the consensus mechanism.

S Sigmoid Transition Shape

Sigmoid regression on $T=10$ PermG detection rates (13 α levels) yields $\alpha_c = 0.483 \pm 0.006$ and $k = 478 \pm 475$. The steepness parameter k has $\mathrm{SE}(k) \approx k$, indicating that the data cannot distinguish among transition shapes: sigmoid, probit, and generalized logistic models all achieve identical negative log-likelihood (13.496). The detection onset is effectively a step function—the 13-point α grid has only one interior transition (0/8 at α=0.45 to 6/8 at α=0.50), providing no information about the transition's functional form. The α_c estimate is stable (SE=0.006) because it is anchored by the step location; the shape parameters are not. We do not fit a combined $T=10+T=20$ sigmoid because α=0.45 yields contradictory counts under the two sequence lengths (0/8 vs. 5/8), reflecting genuine power dependence on T rather than a single dose-response curve.

T Intervention Threshold Sensitivity

Within-α sensitivity at α=1.0. Table 24 reports the predictive cessation outcome under five entropy-drop thresholds (5%–25%) at α=1.0 (Pythia-410M, seed 42). All five thresholds produce identical outcomes: token entropy drops 40.7% at generation 1, far exceeding even the most conservative threshold (25%), so every threshold triggers at the same generation. The intervention conclusion—peak perplexity 57.0, recovery in one generation, final perplexity 21.3—is invariant to threshold choice within this range.

Cross-α operating band. The within-α analysis establishes threshold invariance at α=1.0 but does not address whether thresholds generalize across contamination levels. We examine observed entropy trajectories from the alpha sweep (not actual intervention experiments at α<1.0) to characterize the operating band. The maximum entropy drop across 10 generations varies by contamination level: 7.5% at α=0.0 (fine-tuning drift only), 7.7% at α=0.10, 7.8% at α=0.25, 22.4% at α=0.50, 35.3% at α=0.75, and 62.1% at α=1.0. Thresholds below 7.5% (e.g., 5%) trigger on clean training data—a false alarm. Thresholds above 22.4% (e.g., 25%) miss α=0.50, the lowest contamination level with meaningful entropy signal. The safe operating band is therefore $[10\%, 22\%]$: any threshold in this range produces zero false alarms on α=0.0 while detecting contamination at $\alpha \geq 0.50$. The reported 15% threshold sits near the center of this band. This analysis is based on entropy trajectories from non-intervention training runs; whether actual intervention at α<1.0 would produce the same triggering behavior remains unvalidated.

Table 24: Threshold sensitivity at α=1.0 (Pythia-410M, s42, fp32). All thresholds trigger at generation 1 (entropy drop 40.7%≫threshold), producing identical recovery trajectories.

Threshold	Trigger gen	Peak PPL	Final PPL
5%	1	56.99	21.29
10%	1	56.99	21.29
15%	1	56.99	21.29
20%	1	56.99	21.29
25%	1	56.99	21.29

U Channel Effect Size Analysis

First-differencing eliminates all canary→perplexity Granger associations while preserving canary→diversity leads (§5.2). We present three lines of evidence that this asymmetry reflects a genuine temporal structure difference rather than selective power loss from differencing.

Correlation retention parity. First-differencing retains comparable correlation magnitude for both channel types (Table 25). Diversity channels (n=243 pairs across all non-null experiments) retain a median $|r|$ of 0.952 after differencing, while perplexity channels (n=8 pairs; limited sample) retain 1.00. This parity rules out a blanket power-loss explanation: if differencing selectively destroyed perplexity signal, $|r|$ retention would be substantially lower for perplexity than for diversity.

Table 25: First-differencing effect by channel type. $|r|$ retention = diff $|r|$ / naive $|r|$. Despite comparable $|r|$ retention, Granger F differs by an order of magnitude. Perplexity channel has limited sample (n=8); all paired comparisons are from bf16 experiments.

Channel	n	Naive $\|r\|$	Diff $\|r\|$	Retention	Med. F
Diversity	243	0.943	0.921	0.952	67.8
Perplexity	8	0.924	0.900	1.002	11.5

Granger F-statistic asymmetry. Within matched experiments (same model, seed, α, precision), the Granger F-statistic is 20–28× higher for diversity than perplexity channels. At α=0.60 (s42, bf16): diversity mean F=229.4 versus perplexity mean F=8.3 (ratio 27.8×). At α=1.0 reactive (s43, bf16): diversity mean F=236.4 versus perplexity mean F=11.5 (ratio 20.6×). Since $|r|$ retention is comparable across channels, this F-statistic gap cannot be attributed to differencing-induced signal loss; it must reflect a difference in the temporal structure that Granger tests evaluate.

Temporal structure: lagged vs. contemporaneous. The lag distribution of peak differenced cross-correlations reveals the structural difference. For diversity channels, 81% of peak correlations occur at canary-leading lags (lag<0), with a median lag of -1 generation—consistent with the one-generation temporal precedence visible in Figure 2. For perplexity channels, 50% of peak correlations occur at lag=0 (contemporaneous co-movement), with a median lag of 0. Granger causality tests lagged predictive improvement; contemporaneous co-movement produces low F-statistics regardless of signal magnitude, which is the correct behavior for a test designed to detect temporal precedence. The perplexity channel sample is limited (n=8 pairs from bf16 experiments), and these lag distributions should not be generalized to fp32 conditions without further validation.

V Mutual Information Analysis

We estimate the Gaussian mutual information lower bound between token entropy (canary) and each diversity metric across contamination levels, using T=11 data points per series (T=10 for two conditions). The Gaussian estimator $\hat{I}_G = -\frac{1}{2}\log_2(1 - r^2)$, where r is the Pearson correlation, provides a conservative lower bound on MI; we do not report KSG nonparametric estimates due to uncorrected positive bias at T=11. The α=0.05 data point is excluded because it uses an experiment excluded from the dose-response analysis.

Table 26 reports the mean Gaussian MI across the three distinct-n metrics for each α level. MI follows a V-shaped curve: it drops from 0.112 bits at α=0.10 to a minimum of 0.010 bits at α=0.25, then rises monotonically to 0.270 bits at α=0.75. MI remains positive throughout the blind zone ($\alpha \leq 0.45$; range 0.010–0.141 bits), confirming that canary–diversity coupling exists below the detection threshold but falls below PermG's statistical power floor at T=10. There is no discontinuity at $\alpha_c \approx 0.50$: the detection onset reflects power accumulation, not an abrupt change in information coupling. Bootstrap 95% CIs span 1–3 orders of magnitude at T=11, so individual point estimates should be interpreted cautiously.

Table 26: Gaussian MI lower bound (bits) between token entropy and distinct-n diversity metrics (T=11, Pythia-410M, fp32). Mean across distinct-1/2/3; α=0.05 excluded (see text). CIs omitted due to width at T=11.

α	Seeds	MI (bits)
0.10	2	0.112
0.25	3	0.010
0.30	1	0.062
0.35	1	0.098
0.40	1	0.119
0.45	2	0.141
0.50	3	0.107
0.60	2	0.209
0.75	1	0.270

W Moment Cascade Analysis

We analyze the temporal ordering of metric degradation onset across 6 WikiText seed×α conditions (2 seeds × 3 non-trivial α levels each) to test whether token entropy consistently degrades before diversity metrics. Onset is defined as the first generation where a metric's cumulative change exceeds 5% of its generation-0 value.

Table 27 reports the onset generation for each metric across conditions. Token entropy reaches onset at generation 1 in all 6 conditions; distinct-n metrics reach onset at generation 2 in all 18 pairwise comparisons (6 conditions × 3 distinct-n metrics). Across these 18 comparisons, entropy degrades first in 16 cases (Kendall τ=0.337, consistent with a weak but reliable temporal ordering).

The cascade ordering provides evidence for the

Table 27: Degradation onset generation (5% threshold) by metric and condition (Pythia-410M, WikiText, fp32). Token entropy reaches onset at generation 1 in all non-null conditions; distinct-n metrics follow at generation 2.

Seed	α	Entropy	Dist-1	Dist-2	Dist-3	ECE	PPL
42	0.10	1	2	2	2	2	—
42	0.50	1	2	2	2	1	4
42	0.75	1	2	2	2	1	1
43	0.25	1	2	2	2	4	—
43	0.50	1	2	2	2	1	4
43	0.75	1	2	2	2	1	1

—: onset not reached within T=10 generations.

canary mechanism independent of Granger-type tests: it does not depend on first-differencing, lag selection, or the permutation null distribution. ECE shows onset at generation 1 at $\alpha \geq 0.50$ (both seeds), consistent with its role as a secondary canary metric. At low contamination (α=0.10 and α=0.25), perplexity never reaches the 5% onset threshold within T=10 generations, while entropy already shows generation-1 onset—consistent with entropy's higher sensitivity to distributional shifts.

X Cascade Ordering Theory: Proofs

This section provides the full proofs for Proposition 1 and Proposition 2 from §3, plus supporting results.

Assumption 4 (Mode Amplification)**.** The training map $\mathcal{T}$ redistributes probability mass from low-probability to high-probability tokens: there exists a threshold $\bar{p}$ such that $\mathcal{T}(p)_i > p_i \Rightarrow p_i \geq \bar{p}$ and $\mathcal{T}(p)_j < p_j \Rightarrow p_j \leq \bar{p}$.

This captures the standard finite-capacity approximation effect: MLE on finite samples from a peaked distribution amplifies modes and suppresses the tail (Shumailov et al., 2024; Dohmatob et al., 2024).

Proposition 5 (Entropy is a first-order indicator)**.** *Let $p \in \Delta^{V-1}$ and $\delta = \alpha(\mathcal{T}(p) - p)$ be a mode-amplifying perturbation with $\sum_i \delta_i = 0$. Then $\Delta H = H(p+\delta) - H(p) = -\sum_i \delta_i \log p_i + O(\|\delta\|^2) < 0$.*

Proof. Taylor-expanding H around p: $H(p+\delta) = H(p) - \sum_i \delta_i \log p_i - \sum_i \delta_i - \frac{1}{2}\sum_i \delta_i^2/p_i + O(\|\delta\|^3)$. Since $\sum_i \delta_i = 0$: $\Delta H = -\sum_i \delta_i \log p_i - \frac{1}{2}\sum_i \delta_i^2/p_i + O(\|\delta\|^3)$. Under mode amplification, tokens gaining mass ($\delta_i > 0$) have $p_i \geq \bar{p}$ (hence $\log p_i \geq \log \bar{p}$) and tokens losing mass ($\delta_j < 0$) have $p_j \leq \bar{p}$ (hence $\log p_j \leq \log \bar{p}$). Writing $m = \sum_{i:\delta_i>0} \delta_i$: $-\sum_i \delta_i \log p_i = -m[\overline{\log p}^+ - \overline{\log p}^-] < 0$, where $\overline{\log p}^{\pm}$ are the $|\delta|$-weighted means over gaining/losing tokens. The second-order Fisher information term $-\frac{1}{2}\sum_i \delta_i^2/p_i$ is always negative, reinforcing the decrease. □

Proposition 6 (Diversity is a threshold indicator)**.** *Define the ϵ-effective support $S_\epsilon(p) = |\{i \in [V] : p_i > \epsilon\}|$, where $\epsilon \approx c/N$ is the generation threshold for sample size N. Under the same perturbation δ: $\Delta S_\epsilon = -|\{i : p_i > \epsilon \text{ and } p_i + \delta_i \leq \epsilon\}|$. This equals zero whenever $\|\delta\|_\infty < \gamma(p,\epsilon) := \min_{i:p_i>\epsilon}(p_i - \epsilon)$.*

Proof. Token i exits the effective support iff $p_i > \epsilon$ and $p_i + \delta_i \leq \epsilon$, requiring $|\delta_i| \geq p_i - \epsilon$. If $\|\delta\|_\infty < \gamma$, no token satisfies this condition. □

Proof of Proposition 1. *Part (a):* Take any mode-amplifying δ with $\|\delta\|_\infty < \gamma(p,\epsilon)$. By Proposition 6, $\Delta S_\epsilon = 0$. By Proposition 5, $\Delta H < 0$. Such δ exists for all $\alpha < \gamma / \max_i |\mathcal{T}(p)_i - p_i|$. *Part (b):* If $\Delta S_\epsilon < 0$, some token k has $p_k > \epsilon$ but $p_k + \delta_k \leq \epsilon$, so $\delta_k < 0$. Under mode amplification, this implies $p_k \leq \bar{p}$; the total mass transferred to high-probability tokens ensures $\overline{\log p}^+ > \overline{\log p}^-$. By Proposition 5, $\Delta H < 0$. □

Proof of Proposition 2. By the margin condition, at every generation $s \leq t_H$ the perturbation norm $\alpha\|\mathcal{T}(p_s) - p_s\|_\infty$ is less than $\gamma(p_s,\epsilon)$. By Proposition 6, no token exits the effective support at any $s \leq t_H$, so $\Delta S_\epsilon = 0$ for all $s \leq t_H$. Since time is discrete, $t_D \geq t_H + 1$. □

ECE co-movement (qualitative argument). Under conditional mode amplification (peakier $q_\theta(\cdot|x)$ at each generation), per-token conditional entropy $h(x) = H(q_\theta(\cdot|x))$ decreases while confidence $c(x) = \max_y q_\theta(y|x) \geq \exp(-h(x))$ increases. Averaging: $\bar{H}_t$ decreases, C_t increases. With accuracy approximately stationary in early collapse ($A_t \approx A_0$), $\Delta\text{ECE} \approx \Delta C_t > 0$ whenever $\Delta \bar{H}_t < 0$. This argument establishes the direction of co-movement; the quantitative relationship depends on distribution geometry. $\bar{H}_t$ denotes mean conditional entropy $\mathbb{E}_{x\sim p_{\text{eval}}}[h(x)]$, which differs from the marginal entropy $H_t = -\sum_i p_t(i) \log p_t(i)$ used in Proposition 1; under mode amplification, both decrease monotonically, so the cascade ordering is consistent across both formulations.

MAUVE ordering (conjecture). MAUVE measures divergence in a discretized embedding space via k-means quantization. MAUVE onset requires token-level shifts (detected by entropy) to propagate through n-gram distributions (detected by distinct-n) to aggregate embedding distributions. Empirically, MAUVE onset lags diversity by 0–1 generations on WikiText; however, on C4, MAUVE onset occurs at generation 1 (simultaneous with entropy), suggesting this lag is corpus-dependent.

Empirical support. Among 15 experiments at $\alpha \geq 0.30$ (where entropy perturbation exceeds the detection threshold), cascade ordering $t_H \leq t_D$ holds in 14/15 cases (GPT-2-medium s43 fp32 exhibits bidirectional Granger significance). Table 28 reports onset generations for representative conditions, showing uniform Phase 1/2/3 ordering. Five low-α experiments ($\alpha \leq 0.25$) show entropy perturbation below the detection threshold (gen-1 deviation $< 2\sigma$); cascade ordering is neither confirmed nor refuted in this regime.

Table 28: Cascade onset generations (5% threshold) for representative experiments (Pythia-410M, WikiText, fp32). Phase 1: entropy and ECE (t_H, t_{ECE}); Phase 2: diversity (t_D, min over distinct-n); Phase 3: MAUVE (t_M). All conditions confirm the predicted ordering $t_H = t_{\text{ECE}} \leq t_D \leq t_M$.

Seed	α	t_H	t_{ECE}	t_D	t_M
42	0.50	1	1	2	3
43	0.50	1	1	2	2
42	0.75	1	1	2	3
43	0.75	1	1	2	2

Proof of Proposition 3. At the detection threshold, $\text{Power}(\alpha_c, T) = \pi^*$. Since α_c lies below the non-monotonic dip region ($\alpha_c < 0.55$), the signal model applies directly. From the power function $\text{Power} = \Phi(S(\alpha) \cdot \sqrt{T}/\sigma)$, setting $S(\alpha_c) \cdot \sqrt{T}/\sigma = z^*$ and substituting the signal model $S(\alpha) = c\,\alpha^\beta$ yields $c\,\alpha_c^\beta = z^*\sigma/\sqrt{T}$. Solving: $\alpha_c(T) = (z^*\sigma/c)^{1/\beta} \cdot T^{-1/(2\beta)}$, giving the stated form with exponent $\gamma = 1/(2\beta)$.

Calibration and consistency. Cross-T validation avoids the circularity of fitting theory to model-extrapolated points. We fit $P(\text{det}|\alpha, T) = \Phi(A \cdot \alpha^\beta \cdot \sqrt{T} - z_0)$ to training data at $T \in \{10, 20\}$ via profile likelihood over (β, α_∞), with α_∞=0 emerging as the optimum. The held-out test set ($T \in \{5, 15\}$) yields MAE=0.106 with 95% CI coverage $6/6$: the model correctly predicts all $0/8$ at T=5 and detection at T=15 (predicted 0.49–0.71 vs. observed $6/8$=0.75; under-prediction reflects the $P_{\max}$=0.75 ceiling from MAUVE saturation, which is modeled separately). A linear signal model (β=1, γ=0.5) is rejected. The sub-linear exponent β=0.24 implies that doubling contamination increases signal by only $2^{0.24} \approx 1.18\times$ (18%); since $\alpha_\infty \approx 0$, any nonzero contamination produces signal, but detection requires sufficient T to amplify it. At T=5, $\alpha_c \geq 1.0$: no contamination level is detectable, consistent with the insufficient statistical power at very short observation windows.

Y T=20 Predict-then-Verify Comparison

Table 29 reports the predict-then-verify results for the T=20 extension experiments. Predictions are derived from the detection boundary model (§4.2): detected pairs at T=10 are predicted to remain detected (saturated F-statistics), and undetected pairs are predicted to remain undetected (structural nulls or non-monotonic dip).

Table 29: T=20 predict-then-verify comparison (Pythia-410M, seed 42, WikiText, fp32). Excluding α=1.0, two of four predictions fall within CI (α=0.50, 0.75); boundary cases show systematic bias—α=0.45 falsifies the ceiling hypothesis (under-prediction), α=0.60 over-predicts under raw p-values (v1; matches under FDR v2, not pre-committed).

α	T=10	Predicted T=20	Observed T=20
0.45	0/8	0/8 (ceiling)	5/8
0.50	6/8	6/8 (saturated)	6/8
0.60	0/8	3/8 (v1 raw)†	0/8
0.75	6/8	6/8 (saturated)	6/8
1.00	6/8	6/8 (saturated)	6/8

†V2 (BH–FDR corrected, created 44 min pre-analysis, not git-committed) pred

At $\alpha \geq 0.50$, consistency across T=10 and T=20 supports the detection boundary prediction: PermG counts are determined by signal structure, not observation length. Six diversity-based pairs show F=449–2947 at T=20 (all $\gg F_{\text{crit}}$); two MAUVE-based pairs show F<2 regardless of T, driven by floor saturation (μ=0.029, 100% <0.05) rather than high variance (CV_{MAUVE}=0.089 vs $\text{CV}_{\text{distinct-1}}$=0.139); SNR: MAUVE 4.68 vs diversity 6.40–8.20 (p<0.001). The sole exception is α=0.45 at T=20 (5/8), consistent with boundary proximity. At α=0.45, the shift from 0/8 to 5/8 falsifies Model A (ceiling: α=0.45 structurally undetectable) in favor of Model B (power-limited: $\sqrt{T}$ boost crosses the detection threshold), lower-

ing α_c below 0.45. The α=0.60 non-monotonic dip persists at T=20 (s42: 0/8; s43: 3/8) but dissolves at T=25 (5/8), confirming that $\sqrt{T}$ power scaling eventually overcomes the dip; alternative methods detect signal throughout (DiffXCorr 6/8, TY 5/8 at T=20). PermG directional selectivity is consistent across tested T (T=10, 20): forward 6/8 versus reverse 0/8 at $\alpha \in \{0.50, 0.75, 1.0\}$.

Z T=20 Prediction Timeline

Two versions of the T=20 prediction file exist, reflecting an evolving statistical protocol:

V1 (git-committed, `bad72cc`). Predictions derived from T=10 PermG results using *raw* p-values. At α=0.60, v1 counted 3/8 pairs with raw p<0.05, predicting 3/8 detections at T=20. The observed result (0/8 FDR-significant) does not match.

V2 (pre-analysis, not committed). A revised file incorporating BH–FDR correction at q=0.05, consistent with the correction applied throughout the paper. Under FDR, α=0.60 drops from 3/8 (raw) to 0/8, matching the observation. V2 was created 44 minutes before the formal T=20 analysis began (filesystem timestamp) but was not git-committed, and therefore lacks the cryptographic verifiability of v1.

Scoring. Primary (v1, committed): 3/5 exact matches ($\alpha \in \{0.50, 0.75, 1.0\}$, all saturated regimes). Excluding α=1.0, two of four predictions fall within CI. Both boundary predictions miss: α=0.45 (predicted 0/8, observed 5/8) falsifies the ceiling hypothesis in favor of $\sqrt{T}$ power scaling; α=0.60 (predicted 3/8, observed 0/8) over-estimates detection due to the non-monotonic dip. Sensitivity (v2, FDR-corrected): 4/5 (α=0.60 matches). The systematic conservative bias at α=0.45 is consistent with the $\sqrt{T}$ power increase predicted by Proposition 3.

Task degradation. Table 30 reports end-of-training task performance (generation 10, Pythia-410M, seed 42, fp32) across contamination levels.

At α=0.05, perplexity improves over the uncontaminated baseline (20.8 vs. 26.6), consistent with a mild data augmentation effect. The sharpest degradation occurs between α=0.50 and 0.52 (+31% perplexity), aligning with the PermG detection onset (§5.2).

Table 30: Task degradation across contamination levels (Pythia-410M, seed 42, fp32, generation 10). Perplexity evaluated on held-out WikiText-103.

α	Perplexity	Next-token acc.
Baseline	26.6	0.390
0.05	20.8	0.431
0.25	22.9	0.424
0.46	31.6	0.400
0.50	31.3	0.399
0.52	41.1	0.389
0.60	53.5	0.375
0.75	80.3	0.354